

28th International Conference on Knowledge-Based and Intelligent Information & Engineering Systems (KES 2024)

Trends in consumer evaluations of tuna: Text mining of online reviews

Terumasa Taka^{a*}, Kazuhiko Tsuda^a

^aUniversity of Tsukuba, 3-29-1 Otsuka Bunkyo-ku, Tokyo, 112-0012, Japan

Abstract

This study explores consumer evaluations of Pacific and Southern Bluefin Tuna parts in Japan, using text mining on Rakuten Ichiba reviews from January 2015 to December 2019. Our analysis identified distinct preferences and concerns regarding texture, fat content, and culinary applications, emphasizing species differences. Texture issues, notably sinews in Southern Bluefin Tuna's Akami and unexpected bone in its Chu-toro, highlight the need for improved product descriptions and customer education. Findings imply implications for seafood industry strategies, emphasizing accurate product information for enhanced customer satisfaction. Future research should gather demographic data for tailored strategies. This study demonstrates text mining's role in guiding targeted marketing and refining product in the seafood sector.

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Peer-review under responsibility of the scientific committee of the 28th International Conference on Knowledge Based and Intelligent information and Engineering Systems

Keywords: text mining; correspondence analysis; product review analytics; e-commerce of seafood;

1. Introduction

Tuna ranks among the most widely consumed seafoods globally, with Japan exhibiting vigorous consumption. In 2021, Japan's domestic supply of Tuna was estimated at 340,000 tons [1]. Within Japan, Pacific Bluefin Tuna holds a distinguished status as the pinnacle of quality among tuna varieties. Moreover, Southern Bluefin Tuna, boasting meat

* Corresponding author. Tel.: +81-3-3942-6918.

E-mail address: s2440409@u.tsukuba.ac.jp

quality comparable to Pacific Bluefin Tuna, is sourced from Australia and imported into Japan. These two species are considered high-end tuna in Japan, frequently used for sashimi and sushi preparations in restaurants [2].

Tuna musculature is classified into two categories: normal muscles and dark muscles. The normal muscles of tuna are usually divided into three parts for consumption: O-toro (extremely fatty meat), Chu-toro (moderately fatty meat), and Akami (lean meat). These distinctions are based on their location, color, and mainly their fat content [3, 4].

Chemical analyses have been conducted on the O-toro, Chu-toro, and Akami of Pacific and Southern Bluefin Tuna [5]. In the industry, differences in skeletal structure and meat quality between Pacific and Southern Bluefin Tuna have been noted, such as the tendency of the rib bones of Southern Bluefin Tuna to penetrate the meat [6]. However, few studies have quantitatively analyzed differences in consumer preferences and evaluations in free-text data for each specific part or species of fish.

E-commerce has experienced widespread adoption since the Internet's popularization, emerging as a trend in modern business development [7]. Within the realm of e-commerce, product reviews serve as consumers' feedback consumers, aiding prospective customers in their purchase decisions while also facilitating service improvement efforts [8]. In the food industry, understanding consumer preferences and tastes through review analysis holds significant utility for informing product development and shaping marketing strategy [9].

This study aims to identify trends in consumer evaluations of various fish species and parts by collecting and analyzing product reviews posted by consumers on e-commerce sites regarding O-toro, Chu-toro, and Akami of Pacific and Southern Bluefin Tuna. Leveraging text mining techniques to analyze the characteristics of each fish part and discern trends in consumer evaluations based on commonly recurring words and expressions in the reviews, we seek to provide actionable insights for tailoring marketing strategies and refining product offerings, stratified by fish species and parts.

2. Related work

This chapter discusses previous studies closely related to this research theme, encompassing studies into consumer evaluation using text mining within the fresh food sector, inquiries of evaluation trends regarding fresh seafood, and summarizes the characteristics of this research.

2.1. Research on consumer evaluation using text mining in the fresh food sector

Advances in text mining technology have facilitated the extraction of trends in consumers' food ratings and preferences from online reviews. The analysis of such consumer reviews on food products provides important insights that directly inform products and services improvements. For instance, Hattori et al. analyzed product reviews on e-commerce sites regarding various parts of beef, including by-products [10]. Their findings revealed that consumers expressed concerns not only about the quality, price, and quantity of beef as a whole, but also about aspects such as about order, delivery, and the distinct attributes of different beef cuts, such as tenderness, fat, and price. Another study used text mining to analyze agricultural products, focusing on identifying words with high occurrence rates during the examination of text mining analysis procedures for vegetables [11]. Additionally, Isojima analyzed consumer opinions submitted to consumer cooperatives, revealing variations in evaluations of agricultural products based on their characteristics and seasonal differences [12].

The application of text mining techniques in analyzing consumer reviews across different fresh food sectors, including beef and vegetables, provides a comprehensive understanding of consumer preferences and behaviors. These studies underscore the significant impact of online consumer feedback on product development and marketing strategies. Through systematic extraction and analysis of consumer sentiments, businesses can better tailor their products and services to align with consumer demands and expectations. Furthermore, this approach allows the identification of market trends and potential avenues for service improvement, thereby contributing the growth of the fresh food industry.

2.2. Research on trends in the evaluation of fresh seafood

In the realm of evaluating trends for fresh seafood, traditional approaches have mainly relied on sensory testing and questionnaires. For instance, Xing et al. conducted preference-based sensory evaluations of red sea bream, considering factors such as fish part, gender, and country of origin of the testers, while also examining aspects pertinent to product marketing [13]. Ko et al. used sensory testing to evaluate flatfish and bigeye flounder, aiming to determine the characteristics that contribute to these fish species high ratings [14]. Additionally, Kubo et al. examined the fat content and coloration of sashimi presented at a tuna competition, alongside the evaluation trends observed among the judges [15].

These methods, including sensory evaluations and questionnaires, provide an opportunity to assess fresh seafood products comprehensively. They allow for the acquisition of feedback on sensory attributes such as quality, taste, and texture, while providing valuable insights into consumer purchase motivations and overall satisfaction with the product. Through these methodologies, important insights into consumer evaluations of fresh seafood products have been gleaned, contributing substantially to the formulation of product development and marketing strategies.

However, the research method involving collecting a substantial volume of product reviews on fresh seafood and subsequent examination of trends in consumer evaluations using text mining remains relatively new in the field of fresh seafood and has yet to be fully utilized. The collection and analysis of authentic consumer voices and detailed feedback through product review analysis on online platforms hold significant promise for identifying consumer evaluation trends. Product reviews often include evaluations derived from firsthand experiences with the product and opinions shaped by interactions among consumers, aspects that may elude capture by sensory inspections or questionnaires. Analysis using text mining holds the potential to delve deeper into this information, allowing a detailed analysis of specific factors of consumer expectations, satisfaction, and dissatisfaction.

2.3. Novelty and contribution of this study

This study aims to analyze consumer reviews of O-toro, Chu-toro, and Akami cuts from Pacific and Southern Bluefin Tuna using text mining techniques. The objective is to identify trends in consumer evaluations for each species and part of the fish. Through this approach, our aim is to deepen our understanding of consumer preferences and to contribute to the formulation of product development and marketing strategies within the fresh seafood market. The novelty and contribution of this research lie in the ability to capture specific trends in consumer evaluations of tuna products through the analysis of product review data, exploring perspectives that have not yet been thoroughly investigated.

3. Material and Method

3.1. Data Summary

This study used the Rakuten Ichiba dataset provided by Rakuten, Inc. through the IDR dataset service of the National Institute of Informatics. Rakuten Ichiba is an e-commerce website operated by the Rakuten Group in Japan. The dataset encompassed both product data and product review data sourced from Rakuten Ichiba. We collected product reviews within the seafood category, focusing specifically on Pacific and Southern Bluefin Tuna sashimi (raw cut fish) products. These reviews were posted between January 2015 and December 2019.

To exclusively assess Pacific and Southern Bluefin tuna sashimi, we filtered out products that were processed, seasoned, or sold in assortments or bundled with other fish species. We focused solely on O-toro, Chu-toro, and Akami, the specific cuts commonly consumed in Japan, among the obtained Pacific and Southern Bluefin Tuna sashimi products for our analysis. The first author determined whether the data under analysis corresponded to O-toro, Chu-toro, or Akami by directly examining each product name and description. This data exclude products sold as assortments of different parts.

Table 1 shows the total number of reviews categorized by fish species and part. In total, 3,255 reviews on sashimi were extracted.

Table 1. Number of reviews

Species and parts of tuna	Number of reviews
Pacific Bluefin Tuna O-toro	264
Pacific Bluefin Tuna Chu-toro	969
Pacific Bluefin Tuna Akami	664
Southern Bluefin Tuna O-toro	50
Southern Bluefin Tuna Chu-toro	480
Southern Bluefin Tuna Akami	828
Total	3,255

3.2 Analytical method

For preprocessing and correspondence analysis of product reviews, we employed the KH Coder (ver. 3. Alpha.13m) [16]. Initially, the review text underwent morphological analysis using MeCab to extract words from Japanese sentences [17]. Subsequently, words with significant occurrences throughout the reviews were extracted, and their characteristics were confirmed.

During morphological analysis, Japanese words not included in the lexicon are automatically split, which can introduce problems for analysis. For example, terms like "takeout" will be split into "take" and "out"; potentially altering or losing their original meaning. To address this issue, we took a proactive approach by identifying specific morphemes to be used. These morphemes were then manually corrected by setting forcibly extracted words within KH Coder. Terms like "Negitoro"(Minced tuna meat), "Temaki"(Hand-rolled sushi), "O-toro", "Chu-toro", "Akami", "Wake-ari"(Imperfect), "Minami-maguro"(South Pacific Tuna) were among those forcibly extracted. The analysis focused on nouns, verbs, and adjectives in Japanese.

A correspondence analysis was conducted to cluster highly similar parts from a two-dimensional scatterplot based on the relationship between frequently occurring words within product reviews [18]. Furthermore, using the Jaccard coefficient index, characteristic words were extracted for each fish species and part. The obtained results were used to examine commonalities and differences among and within fish species and part groups concerning post-purchase evaluation criteria and consumption occasions. The minimum number of words parameter was set to 100 as the predefined value for the correspondence analysis in KH Coder.

4. Analysis of results

4.1. Pacific and Southern Bluefin Tuna product review features

Tables 2 and 3 list the top 30 words frequently found in reviews of Pacific and Southern Bluefin Tuna, similar to the study by Hattori et al [10]. These 30 words were classified based on their representation, following the approach of Takezaki et al.[11]. They include words related to quality ("Delicious," "Good," "Taste," "Fat," "Imperfect,"), price ("Price," "Inexpensive"), processing and eating method ("Defrosting," "Sashimi"), emotion ("Think," "Satisfaction," "Pleased"), order and delivery ("Buy," "Repeat," "Order," "Purchase," "Receive"), tuna parts ("O-toro," "Chu-toro," "Akami," "Sinew," "Bone"), and others. Among these, quality (Taste, Fat), and price were considered indicative of the criteria for post-purchase evaluation, reflecting what consumers focused on when evaluating their post-purchase. Furthermore, the consumption context was assumed from the eating method, and the consumption scene (who eats when and how) was also considered to be closely related to post-purchase evaluation. Furthermore, the parts of tuna body were also mentioned, and their influence on the evaluation can be considered.

Table 2. Top 30 words of Pacific Bluefin Tuna reviews

Word	Word count	Word	Word count	Word	Word count
Delicious	1,019	Sinew	249	Pleased	175
Eat	774	Order	249	Great	157
Tuna	641	This time	234	Many	154
Purchase	641	Good	232	Price	153
Defrosting	559	O-toro	208	Contain	149
Think	487	Product	205	Family	148
Chu-toro	359	Buy	205	Repeat	145
Taste	288	Satisfaction	199	Receive	140
Akami	278	Sashimi	184	Imperfect	131
Fat	263	Reach	182	Send	129

Table 3. Top 30 words of South Bluefin Tuna reviews

Word	Word count	Word	Word count	Word	Word count
Delicious	737	This time	190	Bone	130
Eat	479	Order	188	Fat	129
Purchase	448	Product	172	Inexpensive	117
Tuna	411	Good	170	Receive	116
Think	374	Many	168	Repeat	109
Defrosting	314	Contain	166	Family	109
Akami	301	Buy	165	Imperfect	109
Sinew	296	Satisfaction	157	Great	107
Chu-toro	294	Sashimi	138	Reach	107
Taste	240	Price	134	Sinewy	106

4.2. Correspondence analysis

According to the distribution of variables (each tuna species and part) in Fig.1, words common in the reviews of Chu-toro and O-toro appeared in the positive direction for component 1 when viewed from the origin (0, 0). Similarly, words common in the review of Akami appeared in the negative direction. Additionally, many words appeared in the reviews of Southern Bluefin Tuna in the positive direction for component 2 when viewed from the origin (0, 0). Likewise, words more common in reviews of Pacific Bluefin Tuna appeared in the negative direction.

When examining individual species and parts of the fish, words related to sinews notably surfaced in the reviews of Southern Bluefin Tuna Akami. For Chu-toro of Southern Bluefin Tuna, "Bone" emerged as a characteristic word. "Skin" and words related with fat, such as "Toro" and "Fat" were prevalent for O-toro of Southern Bluefin Tuna and Chu-toro and O-toro of Pacific Bluefin Tuna. Words related to seasonal events and gift-giving purposes, such as "New Year," "Father," and "Send," appeared especially in the context of O-toro. Additionally, for Pacific Bluefin Tuna Akami, the word "Donburi"(rice bowl and dishes using rice bowl) appeared frequently.

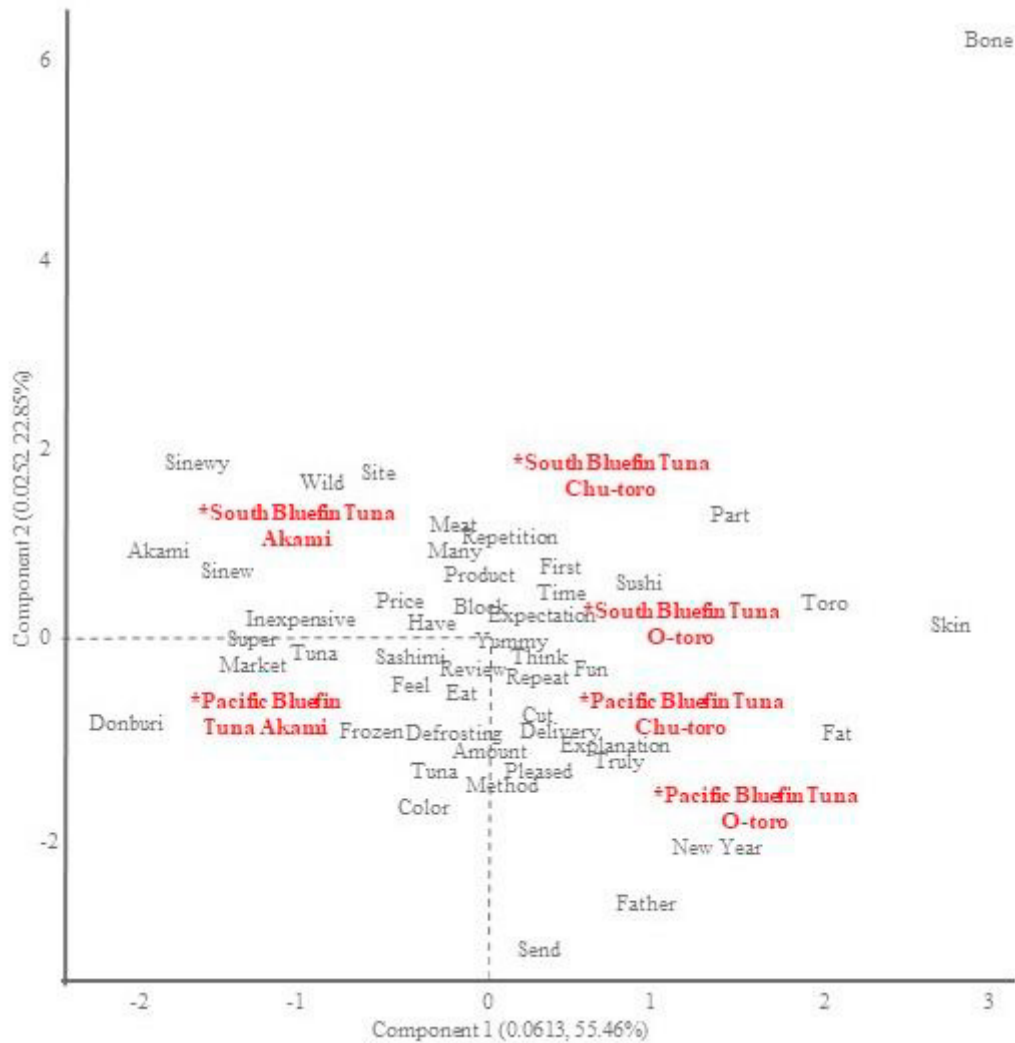


Fig. 1. Result of correspondence analysis

4.3. Discussion of customer evaluation by tuna species and parts

The analysis shows distinct consumer preferences and perceptions for O-toro, Chu-toro, and Akami of both Pacific and Southern Bluefin Tuna. Notably, sinews are more commonly mentioned in connection with Southern Bluefin Tuna's Akami, indicating consumer sensitivity to texture, especially for this species and part. This underscores the importance of texture in consumer satisfaction and suggests a need for targeted communication regarding the textural differences between tuna varieties.

Additionally, the prominent mention of "Donburi" (rice bowl) in reviews of Akami, particularly from Pacific Bluefin Tuna, points to a specific culinary use of this tuna part. This reflects a subtle distinction in the culinary applications of Akami from different species, suggesting that consumers recognize and value different types of tuna for specific culinary contexts. Understanding these subtle distinctions can provide valuable insights for the tuna industry, informing product labeling, marketing strategies, and consumer education efforts to align with the specific preferences and uses identified by consumers.

The term "Bone" was notably more prevalent in reviews for Southern Bluefin Tuna Chu-toro, indicating encounters with unexpected skeletal structures and highlighting areas for both consumer education and product improvement. This finding underscores the necessity not only for customer communications on the website descriptions but also for thoughtful product development efforts aimed at addressing consumer challenges. To mitigate these issues, businesses could consider designing products that minimize bone presence or offer pre-prepared cuts that alleviate preparation concerns. Moreover, while the "Bone" issue serves as a primary example here, it is imperative to adopt a comprehensive approach that addresses other potential consumer concerns identified through review analyses. This could include tackling issues related to texture, taste, or packaging. By integrating consumer feedback into both marketing and product development processes, companies can create more tailored offerings that better meet consumer needs and expectations. Enhancing customer education through detailed product descriptions, cooking tips, and handling instructions on e-commerce platforms can significantly improve the purchasing experience. Together, these strategies can lead to higher satisfaction levels, foster customer loyalty, and drive positive word-of-mouth recommendations.

In the analysis, terms related to fat content, such as "Toro" and "Fat", were predominantly associated with O-toro of Southern Bluefin Tuna and Chu-toro of Pacific Bluefin Tuna, indicating consumer observations or mentions for higher fat content in these varieties. The mention of "Skin" suggests that there might be consumer concerns related to the processing of these tuna parts. This presents an opportunity for e-commerce businesses to enhance consumer guidance on skin removal and preparation techniques or to offer pre-processed, skinless options.

Furthermore, the characteristic association of O-toro with terms related to seasonal events and gift-giving, such as "New Year", "Father", and "Send", reflects its perceived status as a premium product ideal for special occasions. This distinction underscores the need for tailored marketing strategies that highlight the luxurious nature of O-toro, particularly during key gift-giving seasons.

These insights demonstrate that consumer evaluations vary significantly not only between different types of tuna but also among different parts of the same species. This variation emphasizes the importance of nuanced product information and targeted marketing approaches that reflect the specific qualities and recommended uses of each tuna type and part.

While occurrences like unexpected "Bones" in Southern Bluefin Tuna Chu-toro highlight the need for clearer product descriptions and enhanced customer education, it is crucial to recognize that these are not solely issues of consumer misunderstanding. The recurring issues with bones and concerns regarding "Skin" processing underline a significant responsibility for businesses to provide detailed and accurate product information. This responsibility extends beyond basic product descriptions to include comprehensive guidance on preparation and handling, ensuring consumers are well-informed prior to making a purchase. Addressing these concerns through improved communication can mitigate potential dissatisfaction and ensure that consumer expectations are aligned with the realities of the product. As suggested by reviewer feedback, it is essential for businesses to acknowledge and address these information gaps as integral components of their continuous endeavors to enhance consumer satisfaction and trust in their products. Effective communication and accurate information dissemination represent crucial aspect of consumer support, essential not only for fostering satisfaction but also for ensuring safety and enhancing the overall consumer experience.

5. Conclusion

In conclusion, this study has provided insights into consumer evaluations of Pacific and Southern Bluefin Tuna, emphasizing how different parts of the fish—O-toro, Chu-toro, and Akami—are perceived in relation to their texture, fat content, and culinary uses. The distinct mentions of "Sinews" in Southern Bluefin Tuna's Akami and the particular association of "Donburi" with Pacific Bluefin Tuna's Akami highlight nuanced consumer perceptions influenced by texture and culinary application, respectively. These findings underscore the importance of texture and culinary context in driving consumer satisfaction and preference. They suggest that these factors should be carefully considered in crafting marketing and product development strategies.

Moreover, the findings reveal a significant responsibility on the part of businesses to provide accurate and comprehensive product information. This extends beyond basic descriptions to include detailed guidance on product handling and preparation. This is crucial for aligning consumer expectations with the realities of the products and

ensuring a satisfying purchasing experience. As observed, improved communication and comprehensive customer education can mitigate misunderstandings and enhance consumer satisfaction, ultimately fostering loyalty and positive word-of-mouth.

Additionally, the analysis highlights the importance of tailored marketing strategies that leverage the luxury appeal of products like O-toro, particularly during peak gift-giving seasons. This approach acknowledges their status as premium products ideal for special occasions.

However, a limitation of this study is the absence of detailed demographic information about the reviewers, which hinders a more profound understanding of how different consumer segments may perceive the tuna products. Future research should prioritize collecting and analyzing such data to provide more tailored and effective consumer insights.

In summary, this study illustrates the significance of using text mining techniques to analyze consumer reviews, offering valuable insights that can guide more nuanced and effective product development and marketing strategies within the seafood industry. By understanding and addressing the specific qualities and consumer perceptions associated with each tuna type and part of tuna, stakeholders within the tuna industry can better tailor their offerings to meet consumer expectations. This alignment can lead to improved customer satisfaction and has the potential to stimulate increased market demand for tuna products. Looking ahead, it is essential for businesses to embrace their role in providing detailed and accurate product information to enhance the consumer experience and trust in their products.

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